Hossein Mirsaeidi

M.Sc. Student, Aerospace Engineering (Dynamics and Control)
Sharif University Of Technology, Tehran, Iran
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RESEARCH INTERESTS

Control Theory, Safe Control, Multi-Agent Systems, Robotics, and Intelligent Systems.

EDUCATION

Sharif University Of Technology, Tehran, Iran — Sep 2024 – Present
Master of Science in Aerospace Engineering, Majoring in Dynamics and Control — **Cumulative GPA: 19.16/20**
Thesis: "*Safe Intelligent Deadlock Resolution in Cluttered Environments for Multi-Agent Systems*"

Sharif University Of Technology, Tehran, Iran — Sep 2020 – Sep 2024
Bachelor of Science in Aerospace Engineering — **Cumulative GPA: 18.70/20**
Thesis: "*Hardware-in-the-loop Simulation for a Swarm of Autonomous Drones with Obstacle Avoidance Capability*"

National Organization for Development of Exceptional Talents, Isfahan, Iran — Sep 2017 – May 2020
High School Diploma, Mathematics — Cumulative GPA: 19.54/20

ACADEMIC EXPERIENCE

Teaching Assistant:

Department of Aerospace Engineering, Sharif University Of Technology — *Tehran, Iran*

- Advanced Flight Dynamics — Instructor: Dr. Alireza Sharifi — Fall 2025
- Automatic Control ($\times 2$)— Instructor: Dr. Alireza Sharifi — Spring 2023 — Spring 2026
- Control Systems Lab. ($\times 4$) — Instructor: Dr. Alireza Sharifi — Spring 2023 — Summer & Fall 2024 — Fall 2025
- Dynamics ($\times 3$) — Instructor: Dr. Alireza Sharifi — Fall 2022 — Fall 2023 — Fall 2025
- Orbital mechanics — Instructor: Dr. Maryam Kiani — Spring 2023
- Strength of materials — Instructor: Dr. Hossein M.Navazi — Fall 2022

Graduate Research Assistant:

Department of Aerospace Engineering, CNAV Lab, Sharif University of Technology — *Tehran, Iran*

- Implementing quadrotor swarms using Crazyflie platforms. — Present
- Guiding undergraduate students in implementing outdoor drone swarms using PX4. — Fall 2025

PUBLICATIONS

- **H. Mirsaeidi** and A. Sharifi, "Heuristic-Based Trajectory Optimization for Drone Flocking with Obstacle Avoidance," 2023 11th RSI International Conference on Robotics and Mechatronics (ICRoM), Tehran, Iran, 2023, pp. 54-59, doi: 10.1109/ICRoM60803.2023.10412612
- **H. Mirsaeidi**, R. Pordal, and A. Sharifi, "Geometric Deadlock Resolution for Safety-Critical Control in Two-Agent Scenarios," 2025 13th RSI International Conference on Robotics and Mechatronics (ICRoM), Tehran, Iran, November 2025.
- S. Mahfar, **H. Mirsaeidi**, and A. Sharifi, "Applied High-Order Control Barrier Functions for Altitude Control of a Tilt-Rotor Bicopter," 2025 13th RSI International Conference on Robotics and Mechatronics (ICRoM), Tehran, Iran, November 2025.
- **H. Mirsaeidi**, A. Sharifi and R. Pordal, "Kinematic Rotation-Based Deadlock Resolution for Safety-Critical Control of Multi-Agent Systems," submitted to Journal of the Franklin Institute, 2025, under review.

SELECTED PROJECTS

Graduate Projects

- Learning-based deadlock resolution for safe multi-agent coordination: training Soft Actor-Critic policies (Gymnasium/PyTorch) on top of a control-barrier-function safety filter, through a staged curriculum from single-agent navigation to multi-agent intersections. (M.Sc. thesis, ongoing)
- Safe control of quadrotors using real-time control-barrier-function quadratic program, with implementation on Crazyflie 2.1 platforms using Lighthouse positioning.
- Safety-critical control design for a second-order multi-agent system, including graph-theoretic controllability analysis of leader–follower topologies and safety barrier certificates for collision avoidance.
- Adaptive safety-critical control of an uncertain MIMO nonlinear system: adaptive CLF/CBF quadratic programs with concurrent learning, modular ISS-based design (eISS-CLF and ISSf-CBF), under actuation uncertainty.

- Implemented model reference adaptive control (MRAC) for a two-link robotic manipulator with robust adaptive modifications and observer-based output-feedback via loop transfer recovery (LTR).
- Developed and evaluated nonlinear control methods for quadrotor stabilization, including feedback linearization, conventional and super-twisting sliding-mode control, and backstepping augmented with disturbance observers.
- Implemented a sliding-mode controller for aircraft attitude and altitude recovery from flat-spin conditions.
- Implemented machine learning models in Python, including supervised classification (linear/logistic regression, LDA/QDA, SVM), tree-based ensembles, clustering, dimensionality reduction, and deep learning architectures (CNNs, LSTMs, word embeddings) using TensorFlow/Keras.
- Conceptual design of a layered cockpit safety system for unstable-approach protection and automatic go-around, based on fault-tree and human-factors (SHELL) analysis of a case accident.

Undergraduate Projects

- Model-Based Design of a tiltrotor UAV for evaluating guidance, navigation, and control loops.
- Conceptual design of a modern regional jet with a hybrid-hydrogen engine.
- Design and HIL simulation of the PID controller for a 6DoF UAV Quadrotor.
- Implementing Attitude and Heading Reference System using Kalman filter based on a low-cost Inertial Measurement Unit.
- Control design for a ball and beam system.
- Analyzing airline costs and flight routes to identify opportunities for cost reduction.
- Designing a mission scenario for launching a spacecraft to Mars and investigating its feasibility.

SELECTED COURSES

Selected M.Sc. Courses

- **Adaptive Control** — Results pending
- **Principles of Machine Learning** — Results pending
- **Advanced Automatic Control** — Grade: 20/20
- **Nonlinear Systems Analysis (Nonlinear Control)** — Grade: 20/20
- **Advanced Flight Dynamics** — Grade: 18.6/20
- **Navigation & Guidance** — Grade: 20/20

Selected B.Sc. Courses

- **Aircraft Design 1** — Grade: 20/20
- **Automatic Control** — Grade: 20/20
- **Control Systems Laboratory** — Grade: 20/20
- **Instrumentation Workshop** — Grade: 20/20
- **Flight Dynamics 2** — Grade: 20/20
- **Dynamics** — Grade: 20/20
- **Vibrations** — Grade: 20/20
- **Aerial Robotics Design** — Grade: 19/20
- **Foundation Of Automatic Control Design** — Grade: 18.9/20
- **Flight Dynamics 1(Aircraft Performance)** — Grade: 17/20 (Top mark)
- **Orbital Mechanics** — Grade: 20/20

HONORS

- Accepted through the National University Entrance Exam to the Sharif University Of Technology, Department of Aerospace Engineering, and ranked among the Top 0.7% of approximately 150,000 participants. (2020)
- Ranked 4th out of 73 students in the 2020 Aerospace Engineering undergraduate degree. (2024)
- 5th rank in the National MSc. Entrance Exam in Aerospace Engineering of Iranian Universities. (2024)
- Admitted to the Master's program in Aerospace Engineering under the Exceptional Talent Students scheme. (2024)

SKILLS

- **Programming:** MATLAB(Advanced), Python(Intermediate), C(Basic)
- **Simulation & Design:** MATLAB(Simulink & Simscape), Systems Tool Kit, Solidworks, Ansys(Mechanical & Fluent).
- **Hardware**: Crazyflie platform, Raspberry Pi, Arduino, PX4 firmware, QGroundControl.
- **General**: Windows, Linux, Microsoft Office, Git, LaTeX
- **Other Skills:** Model-Based Design, Teaching, Problem-solving, Team Leadership.

LANGUAGES

- Persian (Native)
- English (Advanced, TOEFL iBT: 95/120 – Reading 24, Listening 26, Speaking 23, Writing 22)
- Arabic (Beginner)

REFERENCES

- **Dr. Alireza Sharifi** (Assistant Professor)
  Department of Aerospace Engineering, Sharif University of Technology, Tehran, Iran
  Website: ae.sharif.edu/~portal/faculty/1730782165
  Email: ar.sharifi@sharif.edu
  Phone: +982166168115
  Note: Dr. Sharifi supervised my master's thesis and can provide detailed insights into my research abilities.
- **Dr. Afshin Banazadeh** (Professor)
  Department of Aerospace Engineering, Sharif University of Technology, Tehran, Iran
  Website: ae.sharif.edu/~portal/faculty/1014037799
  Email: banazadeh@sharif.edu
  Phone: +982166168108
  Note: Dr. Banazadeh instructed several of my undergraduate courses and can provide an evaluation of my academic performance.
- **Dr. Maryam Kiani** (Associate Professor)
  Department of Aerospace Engineering, Sharif University of Technology, Tehran, Iran
  Website: ae.sharif.edu/~portal/faculty/1195544945
  Email: kiani@sharif.edu
  Phone: +982166168144
  Note: Dr. Kiani instructed one of my undergraduate courses and I also worked as her teaching assistant.